

PROJECT REPORT

Heart Disease Analysis and Visualization Dashboard

Date	10 July 2026
Team ID	SWTID-2026-8141
Project Name	Heart Disease Analysis and Visualization Dashboard

1. INTRODUCTION

1.1 Project Overview

- Heart disease analytics project introduction
- Need for accessible, visual patient and risk-factor data
- Tableau + MySQL + Flask overview

1.2 Purpose

- Centralize scattered heart disease information into one interactive platform
- Benefits for clinicians, researchers, hospital administrators, and patients
- Support faster, data-driven decisions around cardiac risk and care

2. IDEATION PHASE

2.1 Problem Statement

Heart disease information is distributed across multiple sources — clinical reports, hospital records, and regional case data — making it difficult for users, analysts, and organizations to understand patient risk, diagnosis patterns, hospital infrastructure, and treatment cost. There is a need for an interactive platform that consolidates this information into clear, visual dashboards.

2.2 Brainstorming Ideas

Data Analysis

- Heart disease dataset collection
- SQL database creation
- Data preprocessing and cleaning
- Risk-factor correlation analysis

Visualization

- Interactive Tableau dashboard
- Cardiac hospital location map
- Treatment cost comparison charts
- Dynamic filters and dashboard actions

Clinical Insights

- Diagnosis distribution analysis
- Risk-level classification
- Chest pain type and gender comparison
- Age vs. max heart rate analysis

Web Integration

- Tableau Public publishing
- Dashboard embedding
- User-friendly web interface

Idea Prioritization: Analyze Insights → Support Data-Driven Clinical / Care Decisions → Share Insights & Recommendations

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey covers Awareness, Access, Explore, Understand, and Act stages, tracking how a patient or clinician moves from first learning about the platform to acting on the risk insights it surfaces.

3.2 Solution Requirement

Functional Requirements

- Collect and process heart disease datasets from multiple sources.
- Store patient and hospital information using a MySQL database.
- Retrieve and analyze data using SQL queries.
- Create interactive Tableau visualizations for better data understanding.
- Display patient risk factors, diagnosis distribution, and hospital/cost insights.
- Provide dynamic filters and dashboard actions for exploration.

Non-Functional Requirements

- Usability, Performance, Reliability, Availability, Scalability, and Privacy (anonymized data with no personally identifiable clinical records).

3.3 Technical Architecture

The architecture follows a layered approach: Data Layer (CSV datasets) → Processing Layer (Python/Pandas + MySQL/SQL) → Visualization Layer (Tableau Desktop / Tableau Public) → Application Layer (Flask + HTML/CSS/JS) → Presentation Layer (responsive web interface).

3.4 Technology Stack

Database Technology — MySQL: Used for storing and managing heart disease datasets. Helps organize structured data and execute SQL queries for analysis. Tools: MySQL Server 8.0, MySQL Workbench.

Visualization Technology — Tableau: Used for creating interactive dashboards and stories from processed heart disease datasets. Features used: Worksheets, Dashboards, Stories, Filters, Maps, Tableau Public Publishing.

Web Technology — Flask: Used to create a web interface for displaying embedded Tableau dashboards and stories. Technologies: Python, Flask, HTML5, CSS3, Bootstrap, JavaScript.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit explains how the Heart Disease Analytics Platform addresses challenges faced by clinicians, researchers, hospital administrators, and patients in understanding scattered cardiac health information. The solution combines multiple heart disease datasets and provides interactive Tableau dashboards to analyze risk factors, diagnosis patterns, hospital infrastructure, and treatment cost. This helps users make better decisions through centralized and visual data insights.

4.2 Proposed Solution

The proposed solution is a Heart Disease Analytics Platform that integrates MySQL database management, Tableau visualization, and Flask web technology. Patient, hospital, and cost datasets are cleaned and structured, then connected with Tableau Desktop to create interactive dashboards, charts, maps, KPIs, and stories.

The dashboards are published through Tableau Public Server and embedded into a Flask web application using HTML, CSS, and JavaScript. End users such as clinicians, researchers, and patients can access the web application to explore heart disease insights interactively.

5. PROJECT PLANNING & SCHEDULING

Project planning defines the complete development process of the Heart Disease Analytics Platform. Project activities were divided into sprint phases: dataset collection and preparation (Sprint 1), risk-factor and diagnosis visualization (Sprint 2), and treatment-cost analysis, story development, and web integration (Sprint 3). Total estimated story points: 27+; velocity approximately 13-14 points per sprint.

6. TESTING

Functional and performance testing was carried out across the following areas:

- Data rendering of all 4 CSV datasets (clinical, regional, hospital, cost).
- Data preprocessing accuracy — duplicate removal, missing-value handling, standardization.
- Interactive filter behavior (age group, gender, risk level, region, chest pain type).
- Dashboard visualizations — Cardiac Hospitals by Region and Type, Hospital Location Map, Diagnosis Distribution, Max Heart Rate by Age Group, Treatment Cost Comparison, Top Risk Factors, Chest Pain Type by Gender, Diagnosis Treemap.

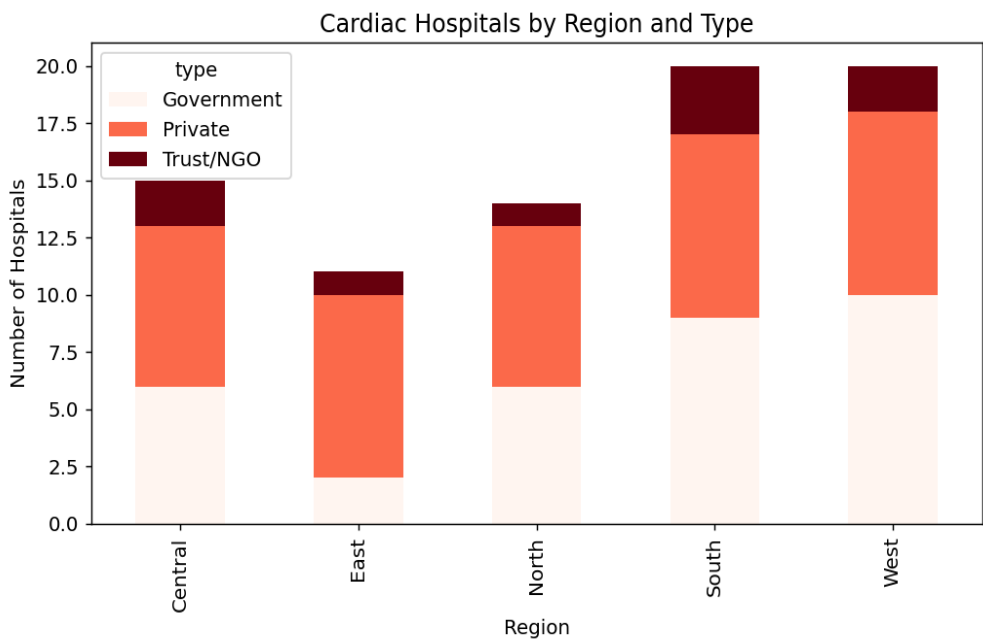
Publishing and Web Integration Testing

The completed Tableau dashboard and story were published successfully using Tableau Public. The Tableau Public embed code was integrated into the Flask web application, allowing users to access interactive heart disease analytics through a web interface. Web page responsiveness and dashboard loading performance were successfully verified.

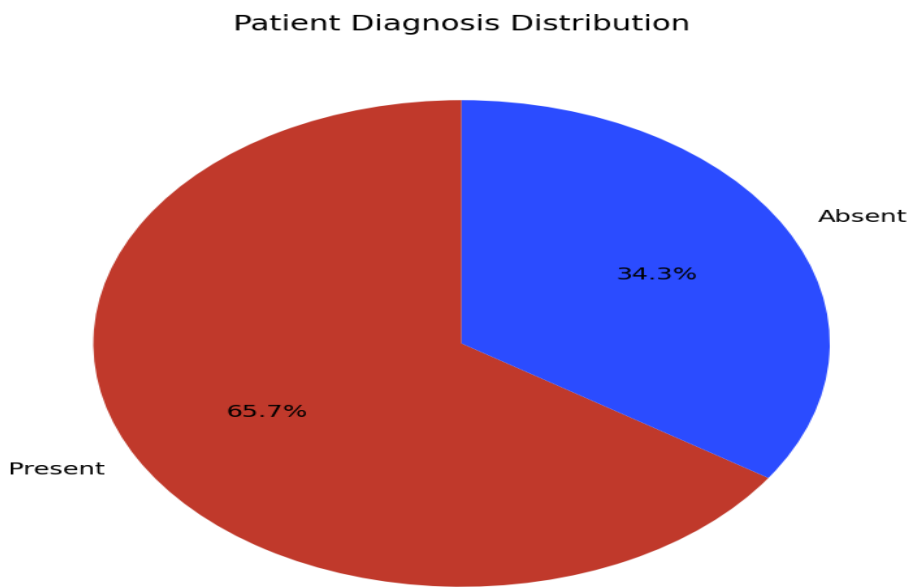
Testing Result: The Heart Disease Analytics Platform successfully passed functional and performance testing.

7. RESULTS / OUTPUT SCREENSHOTS

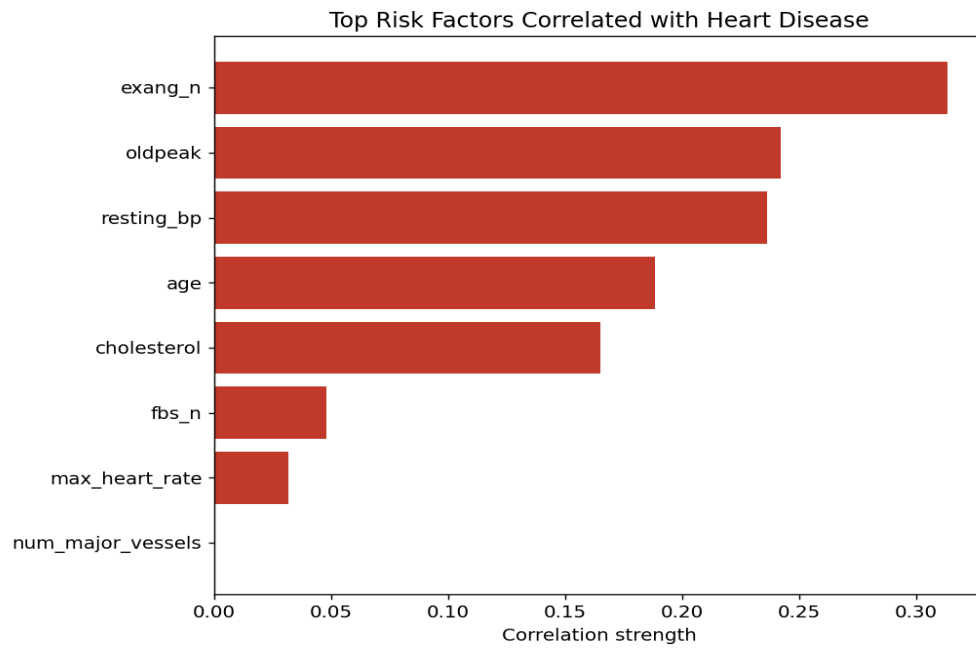
The interactive Tableau dashboard displays complete heart disease analysis including hospital infrastructure, hospital map, diagnosis distribution, heart rate analysis, treatment cost, top risk factors, chest pain type, and diagnosis treemap. Users can interact with dashboard filters to explore specific patient and hospital information.



Cardiac Hospitals by Region and Type



Patient Diagnosis Distribution



Top Risk Factors Correlated with Heart Disease

The Tableau story presents heart disease analytics in a sequential storytelling format, helping users understand important patient and hospital trends step by step. The Flask web application provides a user-friendly interface where Tableau dashboards and stories are embedded for interactive exploration.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides interactive visualization of heart disease data through Tableau dashboards and stories.
- Integrates multiple heart disease datasets into a centralized analytics platform for comprehensive analysis.
- Enables users to compare patients based on risk factors, diagnosis, and treatment cost.
- Offers geographical visualization of cardiac hospital distribution using interactive maps.
- Dynamic filters allow users to explore patient risk levels, chest pain types, and regions.

Disadvantages

- Uses a sample/synthetic dataset rather than a live, real-time clinical feed.
- Not intended for direct medical diagnosis — requires clinician interpretation.
- Dependent on Tableau Public availability for dashboard/story embedding.

9. CONCLUSION

Interactive dashboards and stories were created to represent patient risk factors, diagnosis distribution, cardiac hospital infrastructure, and treatment cost. The Tableau dashboards were published through Tableau Public and integrated into a Flask web application to provide a simple and interactive user experience. The project demonstrates how data analytics and visualization technologies can help clinicians, researchers, and patients understand heart disease trends and support better data-driven decisions.

10. FUTURE SCOPE

- Integration with real, ethically sourced, and properly anonymized clinical datasets.
- Live hospital bed / ICU / cath-lab availability tracking.
- AI-based heart disease risk prediction using Machine Learning.
- Wearable device (ECG / heart rate) data integration.
- Cloud deployment and mobile application support.
- National-level cardiac health analytics and preventive-care planning.

Dataset Note:

This project uses a synthetic, illustrative heart disease dataset generated for educational purposes. Replace with a properly sourced dataset (e.g., the UCI Heart Disease dataset) for research or real submission use.

GitHub & Project Demo Link

GitHub Repository: [Add your GitHub repository link here]

Tableau Dashboard Public Link: [Add your published Tableau dashboard link here]

Tableau Story Public Link: [Add your published Tableau story link here]

Flask Demo Link: [Add your deployment link here]